

IntCoOp: Interpretability-Aware Vision-Language Prompt Tuning

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Abstract

Image-text contrastive models such as CLIP learn transferable and robust representations for zero-shot transfer to a variety of downstream tasks. However, to obtain strong downstream performances, prompts need to be carefully curated, which can be a tedious engineering task. To address the issue of manual prompt engineering, prompt-tuning is used where a set of contextual vectors are learned by leveraging information from the training data. Despite their effectiveness, existing prompt-tuning frameworks often lack interpretability, thus limiting their ability to understand the compositional nature of images. In this work, we first identify that incorporating compositional attributes (e.g., a “green” tree frog) in the design of manual prompts can significantly enhance image-text alignment scores. Building upon this observation, we propose a novel and interpretable prompt-tuning method named **IntCoOp**, which learns to jointly align attribute-level inductive biases and class embeddings during prompt-tuning. To assess the effectiveness of our approach, we evaluate **IntCoOp** across two representative tasks in a few-shot learning setup: generalization to novel classes, and unseen domain shifts. Through extensive experiments across 10 downstream datasets on CLIP, we find that introducing attribute-level inductive biases leads to superior performance against state-of-art prompt tuning frameworks. Notably, in a 16-shot setup, **IntCoOp** improves CoOp by 7.35% in average performance across 10 diverse datasets.

1 Introduction

Recently, significant advancements have been achieved in the field of vision-language models, with notable examples like CLIP (Radford et al., 2021), Flamingo (Alayrac et al., 2022), ALIGN (Jia et al., 2021a), and CoCa (Yu et al., 2022). These models have excelled in acquiring transferable and robust image representations, a feat accomplished

through a combination of two fundamental components: (i) Large-scale paired image-text datasets ranging from 400M to 2B image-text pairs; (ii) A contrastive objective that aligns the image and text embeddings into a common subspace. Leveraging these ingredients, vision-language models have obtained strong performances in zero-shot classification, image-text retrieval, and robustness to distribution shifts. For all these tasks, contrastive models such as CLIP enable zero-shot inference: Given an image $\mathcal{I}$ and a set of text prompts $\{t_i\}_{i=1}^N$, the most relevant text-prompt $t \in \{t_i\}_{i=1}^N$ is identified by maximizing the image-text similarity between $\mathcal{I}$ and t .

Adapting image-text contrastive models for downstream tasks is a complex undertaking. Achieving optimal performance with image-text contrastive models necessitates the manual creation of domain-specific prompts, a process that demands extensive domain knowledge and is exceptionally challenging and time-consuming. Even with considerable prompt engineering, there is limited assurance that the designed prompt is truly optimal. To address this issue, recent research (Zhou et al., 2022a; Lee et al., 2023; Khattak et al., 2023; Ouali et al., 2023) has turned to prompt-tuning techniques, borrowing concepts from the field of NLP and applying them to vision-language models like CLIP to achieve good image recognition performance on downstream tasks. However these frameworks often *lack interpretability* and as a result the model struggles to understand the composition of the images.

In this study, we address this challenge by learning a method to extract and embed attribute-level information into the prompt-tuning framework during training. We define an *attribute* as an interpretable concept that is relevant to the image and encapsulates its semantic essence. Although manually crafted prompts can vary in their characteristics based on the specific downstream domain,

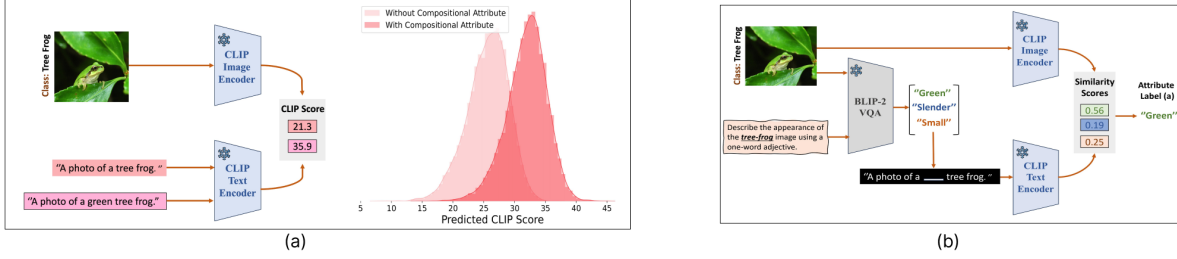


Figure 1: **(a) Importance of learning interpretable concepts in prompts.** **Left:** For each image, we design two prompt templates: (1) Without any compositional attribute “A photo of a $[cls]$ ” and (2) With compositional information “A photo of a $[a]$ $[cls]$ ” where $[cls]$ represents the classname and $[a]$ represents an attribute obtained using a BLIP-2 based VQA model. **Right:** The distribution plot highlights the importance of baking attribute information into the prompts. For this analysis, we used a CLIP model with a ViT-B/16 image encoder and a dataset consisting of 50 images selected randomly from each of 1000 classes in ImageNet-1k. The x-axis indicates the predicted CLIP score. Clearly, the CLIP model is more confident when the prompts include information related to the compositionality of the image. **(b) Framework for obtaining attribute-level supervision.** We present the overarching architecture for generating attribute labels a for a given training image using BLIP-2 VQA model.

our analysis has revealed a noteworthy trend. We have observed that prompts containing attribute information that describes the objects in the images lead to enhanced image-text alignment scores in contrastive models such as CLIP. For instance, as depicted in Figure 1, we can see that prompts incorporating compositional attributes such as “green” tree frog yield higher image-text alignment scores than those lacking such descriptors.

Based on these findings, we present an interpretable prompt-tuning approach known as **IntCoOp**, which incorporates attribute information into the prompt-tuning procedure thereby generating more interpretable prompts. While one might initially consider leveraging off-the-shelf image captioning models to generate attribute labels, this approach becomes infeasible during inference when class labels are unavailable. Consequently, generating attribute descriptions for images emerges as a *non-trivial task*. To mitigate this challenge, we train a compact hypernetwork responsible for predicting embeddings corresponding to attribute descriptors.

We test our prompt-tuning method **IntCoOp** on a range of diverse downstream datasets to test for generalization to novel classes, and robustness to distribution shifts. In Section 5, we show that our method **IntCoOp** has improved robustness to distribution shifts, domain generalization, and few-shot learning. Notably, in domain generalization setup, **IntCoOp** outperforms PLOT (Chen et al., 2023) by 19.32% in average performance across 4 diverse domains. In summary, our research provides compelling empirical support for the substantial advantages of integrating attribute-level inductive biases into the prompt-tuning process.

Overall, our paper makes the following key con-

tributions:

- We introduce a novel prompt-tuning method, named **IntCoOp**, which concurrently aligns attribute-level inductive biases and class embeddings during training, thus facilitating the generation of interpretable prompts.
- We devise an efficient cross-attention mechanism to integrate image information with the learnable prompt tokens seamlessly.
- We present comprehensive experiments across a range of tasks, including generalization to unseen classes, and distribution shifts showing the efficacy of **IntCoOp**. Notably, in a 16-shot setup, **IntCoOp** outperforms the state-of-art framework LFA (Ouali et al., 2023) by 1.27% improvement in average performance across 10 diverse datasets.

2 Related Works

Pretrained Vision-Language Models. Recent research (Radford et al., 2021; Yu et al., 2022) has shown that leveraging language to train image encoders can result in strong downstream performances especially for robustness and few-shot learning. These vision-language models are usually pre-trained on large corpuses of image-text pairs using contrastive objectives that align image and text representations into a common subspace. CLIP (Radford et al., 2021) and ALIGN (Jia et al., 2021b) use *only* the contrastive objective to align image-text embeddings. For example, CLIP is pre-trained on $\sim 400M$ image-text pairs whereas ALIGN is pre-trained on a much larger set of $\sim 1B$ image-text pairs. CoCa (Yu et al., 2022) uses a captioning loss in conjunction with contrastive objectives to further improve image representations.

However, in all these vision-language models, inference requires manually curated prompts to extract the best performance, which can be a tedious engineering task. To mitigate this issue, recent research has turned to prompt-tuning techniques to automatically learn domain specific prompts.

Prompt Tuning. Given a set of text-instructions and an image, existing vision-language models make their decisions by selecting the text-instruction which has the maximum similarity between the image and text-embeddings. Recent advances in this field, such as methods like CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), ProDA (Lu et al., 2022), VPT (Jia et al., 2022), MaPLe (Khattak et al., 2023), KgCoOp (Yao et al., 2023), ProGrad (Zhu et al., 2022), LASP (Bulat and Tzimiropoulos, 2023), RPO (Lee et al., 2023), DAPT (Cho et al., 2023), PLOT (Chen et al., 2023), and LFA (Ouali et al., 2023) have shifted from manually designed prompts to automatically learning prompts through fine-tuning learnable vectors with image-text pairs from the target domain. CoOp fine-tunes CLIP to optimize a set of learnable tokens in the input layer of the text-encoder. CoCoOp enhances CoOp by incorporating conditional image information in the prompt-learning process. VPT learns tokens in each layer of a given encoder through a fine-tuning objective. KgCoOp introduces a regularizer to constrain the prompt tuning process such that the representations of the learned prompts do not deviate significantly from the manually crafted prompts. ProGrad leverages the prompt gradients to fine-tune the learnable tokens such that the prior knowledge in the vision-language model is preserved. PLOT applies optimal transport to match the vision and text modalities for generating the discriminative and visual-aligned local textual prompt tokens. Overall, none of the existing works aim to understand if augmenting certain inductive biases in the prompt-tuning process is beneficial. Our work **IntCoOp** specifically addresses this issue and shows that incorporating compositional attributes in the prompt-tuning process can indeed be beneficial for downstream tasks.

3 Preliminaries

Contrastive Language-Image Pre-training (CLIP) (Radford et al., 2021) is a vision-language model trained on a large dataset of 400 million image-text caption pairs using a contrastive loss. CLIP primarily consists of two

major components:

(1) **Vision Encoder** $\mathcal{V}(\cdot)$ consists of a ViT (Dosovitskiy et al., 2020) model, which takes an image $\mathcal{I} \in \mathbb{R}^{H \times W \times 3}$ as input and outputs a visual embedding in the latent space. The vision encoder $\mathcal{V}$ consists of L transformer blocks $\{\mathcal{V}_i\}_{i=1}^L$. First, the input image $\mathcal{I}$ is split into R fixed-size patches which are projected into patch embeddings $E_0 \in \mathbb{R}^{R \times D_v}$, where D_v is the constant latent vector size of the image encoder. Patch embeddings E_i are input to the $(i + 1)^{\text{th}}$ transformer block ($\mathcal{V}_{i+1}$) along with a learnable class token $\mathbf{x}_i$ and is sequentially processed through L transformer blocks:

$$[\mathbf{x}_i, E_i] = \mathcal{V}_i([\mathbf{x}_{i-1}, E_{i-1}]) \quad i = 1, 2, \dots, L.$$

To obtain the final image representation, the class token $\mathbf{x}_L$ of the last transformer layer ($\mathcal{V}_L$) is projected to a common image-text latent embedding space via a linear projection layer.

$$\mathcal{V}(\mathcal{I}) = \text{Proj}(\mathbf{x}_L) \quad \mathbf{x}_L \in \mathbb{R}^{D_{vl}}.$$

where D_{vl} is the constant vector size of the image-text latent embedding space.

(2) **Text Encoder** $\mathcal{T}(\cdot)$ is a transformer-based model that maps the input text captions into text embeddings.

For zero-shot inference on a downstream dataset consisting of C classes with class names $\{[cls]_c\}_{c=1}^C$, CLIP uses hand-crafted prompts to generate the textual class embeddings. Specifically, given a hand-crafted prompt template “A photo of a $[cls]$ ”, let $\mathbf{s}_c$ represent the sequence embedding for the prompt “A photo of a $[cls]_c$ ” corresponding to the c -th class. Given an input image $\mathcal{I}$, the output probability is given by:

$$\mathbb{P}(\hat{y} = c | \mathcal{I}) = \frac{\exp(\cos(\mathcal{V}(\mathcal{I}), \mathcal{T}(\mathbf{s}_c)) / \tau)}{\sum_{j=1}^C \exp(\cos(\mathcal{V}(\mathcal{I}), \mathcal{T}(\mathbf{s}_j)) / \tau)} \quad (1)$$

where $\cos(\cdot, \cdot)$ represents the cosine similarity and τ is the temperature coefficient.

Context Optimization (CoOp) (Zhou et al., 2022b). Designing hand-crafted prompts in CLIP for every downstream data set is a tedious and time-consuming task. To mitigate this issue of prompt engineering, CoOp (Zhou et al., 2022b) proposed to learn the prompts directly from the data by replacing the hand-crafted prompt with a context vector comprising of M tunable vectors. Let the context vector be represented as

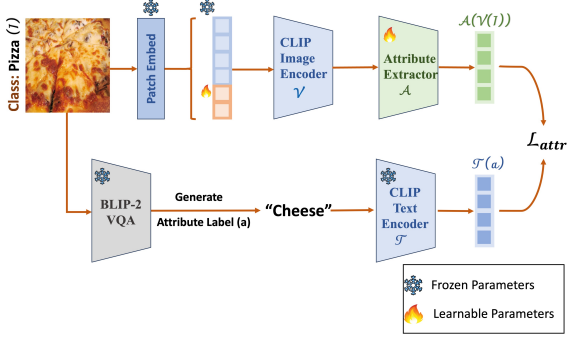


Figure 2: **Framework for learning compositional attributes.** The figure elucidates the training framework of the attribute extractor network $\mathcal{A}$.

$\mathbf{u} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M\}$, where $\mathbf{u}_i$ represents a 512-dimensional vector¹. Unlike the hand-crafted prompt template, the tunable prompts are now designed as $\mathbf{p} = \{[\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M, [cls]_c]\}_{c=1}^C$. To allow the exchange of information learned from the data, the context vector $\mathbf{u}$ is common across all the class categories. Finally, the context vector $\mathbf{u}$ is learned by minimizing the cross-entropy loss between the ground-truth and predicted label as follows:

$$\mathbb{P}(\hat{y} = c | \mathcal{I}) = \frac{\exp(\cos(\mathcal{V}(\mathcal{I}), \mathcal{T}(\mathbf{p}_c)) / \tau)}{\sum_{j=1}^C \exp(\cos(\mathcal{V}(\mathcal{I}), \mathcal{T}(\mathbf{p}_j)) / \tau)} \quad (2)$$

$$\mathcal{L}_{CE} = -\log \mathbb{P}(\hat{y} = y | \mathcal{I}) \quad (3)$$

where, y represents the true label for image $\mathcal{I}$ and $\mathbf{p}_c$ represents the tunable prompt for class c . Note that during training **IntCoOp**, the vision and text encoder in CLIP are completely *frozen* and the optimization framework only updates the context vector $\mathbf{u}$.

4 IntCoOp: Interpretability-Aware Prompt Tuning

In this section, we provide a detailed overview of our proposed prompt-tuning approach **IntCoOp**. In Section 4.1, we detail the process of extracting attribute information from a given image. Next, in Section 4.2, we delve deeper to understand the process of generating image-conditioned prompts. Finally, we outline our entire training framework in Section 4.4, demonstrating the integration of all components into the training pipeline. Similar to past context optimization approaches (Zhou et al., 2022b), **IntCoOp** can also be easily applied to a broad family of CLIP-like vision-language models.

¹The vector $\mathbf{u}_i$ is of same dimension as the word-embedding of class names $[cls]_c$. In this study, we primarily use CLIP-ViT-B/16 model where text embeddings are projected in a 512-dimensional space.

4.1 Learning Interpretable Image Concepts

Obtaining Attribute-level Supervision. Given an input image $\mathcal{I}$, our goal is to extract an interpretable attribute (denoted by a) that provides an accurate characterization of the image. For example, given the image of “Tree Frog” in Figure 1(b), we can define the attribute a as “Green”. However, standard image-recognition datasets such as Imagenet (Deng et al., 2009) only provide true labels for object categories and do not consist of attribute-level supervision. We overcome this problem by using a BLIP-2 (Li et al., 2023) ViT-G FlanT5XXL based VQA model to generate an attribute label ($a_{\mathcal{I}}$) for each image $\mathcal{I}$ in the train set. The entire framework is visually represented in Figure 1(b). We refer the reader to the Appendix B for a detailed description and visualization of more representative examples.

Learning to extract attribute information during training. During inference, as class labels are unavailable for test images, direct utilization of off-the-shelf captioning models (Li et al., 2023) is infeasible. To circumvent this limitation, we propose training a network to learn contextually relevant attributes (see Figure 2). Specifically, we design an attribute extractor network $\mathcal{A}$, which takes as input the image embedding from CLIP’s vision encoder and outputs a 512-dimensional vector representing the embedding of the attribute. This network is trained using supervised attribute labels obtained from the framework in Figure 1(b).

Designing the attribute extractor. It is important to note that the attribute extractor network $\mathcal{A}$ learns the interpretable concepts directly from the image embedding. Therefore, the embedding vector must effectively encode information regarding the compositionality of the image to enable proper training of the network. In Table 3, we show that the embeddings from CLIP’s frozen vision encoder are not expressive enough to essentially capture the attribute information. This challenge is compounded by the fact that, in a few-shot setup, there are a limited number of samples available for each class, leading to suboptimal training of the attribute extractor. To generate richer and more informative visual representations, we append a set of n learnable parameters $\{\mathbf{Z}_i^j \in \mathbb{R}^{D_v}\}_{j=1}^n$ to each transformer layer $\mathcal{V}_i$ of the image encoder up to depth K .

$$[\mathbf{x}_i, E_i, _] = \mathcal{V}_i([\mathbf{x}_{i-1}, E_{i-1}, Z_{i-1}]) \\ \forall i = 1, 2, \dots, K.$$

$$[\mathbf{x}_j, E_j, Z_j] = \mathcal{V}_i([\mathbf{x}_{j-1}, E_{j-1}, Z_{j-1}]) \quad \forall j = K+1, \dots, L.$$

$$\mathcal{V}(\mathcal{I}) = \text{Proj}(\mathbf{x}_L)$$

In Section 7, we show that this improved design choice leads to better performance on downstream tasks. Finally, the generated attribute labels can be used to train the network $\mathcal{A}$ by minimizing the following loss:

$$\mathcal{L}_{\text{attr}} = \|\mathcal{A}(\mathcal{V}(\mathcal{I})) - \mathcal{T}(a_{\mathcal{I}})\|_f^f \quad (4)$$

where $\|\cdot\|_f^f$ indicates the f -th norm, $\mathcal{T}(a_{\mathcal{I}})$ represents the 512-dimensional token embedding of the attribute $a_{\mathcal{I}}$. In Appendix E, based on ablations we find setting $f = 2$ gives the best performance. In this paper, we instantiate the network $\mathcal{A}$ with a two-layer neural net with ReLU activations.

4.2 Instance-Conditional Prompts

In this section, we delve deeper into understanding how the prompts are generated. Recall from Section 3, that for CoOp (Zhou et al., 2022b), the context vector $\mathbf{u} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M\}$ is shared across all classes, and the tunable prompts are designed as $\mathbf{p} = \{[\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M, [cls]_c]\}_{c=1}^C$. In Table 3, we show that sharing the context vectors across all images leads to sub-optimal generalization to novel classes. To address this concern, we opt for a strategy that involves generating instance-conditional context tokens. However, rather than a straightforward addition of the image embedding to the context tokens (Zhou et al., 2022a), we employ a Multi-head Attention module. This module generates context tokens by attending to the image embedding. Given an input image $\mathcal{I}$, the image attended context vector $\mathbf{h}(\mathcal{I})$ is given by:

$$\mathbf{h}(\mathcal{I}) = \text{MultiHead}(\text{Query}=\mathbf{u}, \text{Key}=\mathcal{V}(\mathcal{I}), \text{Value}=\mathcal{V}(\mathcal{I}))$$

where $\mathbf{u}$ represents the context vector, and MultiHead indicates a Multi-head attention module. Note that the instance-conditioned context vector $\mathbf{h}(\mathcal{I})$ has the same shape as $\mathbf{u}$.

Finally, we can generate the prompts for each class by embedding the output of the attribute extractor into the instance-conditioned context vector $\mathbf{h}(\mathcal{I})$. Let $\mathbf{p}^+(\mathcal{I})$ represent the attribute incorporated prompts and is defined as:

$$\mathbf{p}^+(\mathcal{I}) = \{[\mathbf{h}_1, \dots, \mathbf{h}_M, \mathcal{A}(\mathcal{V}(\mathcal{I})), [cls]_c]\}_{c=1}^C \quad (5)$$

Unlike prior works (Zhou et al., 2022a), our cross-attention based image-conditioning mechanism incorporates a learned weighted sum of various points in the image embedding for a single position in the context vector, thereby providing a stronger conditioning signal. In Section 7, we empirically show that our conditioning mechanism is better suited for few-shot fine-tuning in CLIP.

4.3 Regularizing the Prompts

Analysis by Yao et al. (2023) reveal that without any regularization, the context vectors may heavily overfit the training data. This can lead to poor performance on unseen classes during inference. To mitigate this, they propose adding a knowledge-guided loss that aims to minimize the discrepancy between the learned prompts and the handcrafted template ‘‘A photo of a $[cls]$ ’’. In this paper, we also add an additional loss term to regularize the learned prompts. However, instead of simply using the hand-crafted templates, we generate a set of textual prompts incorporating the compositional information for each image. Given an image $\mathcal{I}$, let $\{\mathbf{p}_i^{\text{gen}}(\mathcal{I})\}_{i=1}^N$ represent the pool of N synthetically generated prompt templates embedded with interpretable concepts $a_{\mathcal{I}}$ in image $\mathcal{I}$. In this study, we select $N = 80$ diverse textual prompts as suggested in Radford et al. (2021). Based on this, we define the regularization loss as:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^N \|\mathcal{T}(\mathbf{p}^+(\mathcal{I})_y) - \mathcal{T}(\mathbf{p}_i^{\text{gen}}(\mathcal{I}))\|_g^g \quad (6)$$

where y represents the true label for the image $\mathcal{I}$, $\mathcal{T}(\cdot)$ is the CLIP text encoder and $\mathbf{p}^+(\mathcal{I})_y = [\mathbf{h}_1, \dots, \mathbf{h}_M, \mathcal{A}(\mathcal{V}(\mathcal{I})), [cls]_y]$ is the learnable prompt for the true class y . Based on ablations in Appendix E, we set $g = 1$.

4.4 Putting it together

Let $\mathcal{D}^{\text{train}} = \{\mathcal{I}_j, y_j\}_{j=1}^J$ represent a training dataset consisting of J samples, where $\mathcal{I}_j$ is an image and $y_j \in \{1, \dots, C\}$ represents the corresponding label. Given the dataset, we first generate the attribute labels ($a_{\mathcal{I}}$) for each image as defined in Section 4.1. Note, to avoid any computational overhead during training, we perform this operation offline. Based on the previous discussions, the training loss is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{attr}} + \lambda_2 \mathcal{L}_{\text{reg}} \quad (7)$$

$$\text{where } \mathcal{L}_{\text{CE}} = -\frac{1}{J} \sum_{j=1}^J \log \frac{\exp(\cos(\mathcal{V}(\mathcal{I}_j), \mathcal{T}(\mathbf{p}^+(\mathcal{I}_j)_{y_j}))/\tau)}{\sum_{c=1}^C \exp(\cos(\mathcal{V}(\mathcal{I}_j), \mathcal{T}(\mathbf{p}^+(\mathcal{I}_j)_c))/\tau)}$$

where y_j represents the true label for the image $\mathcal{I}_j$ and C represents the number of seen classes. The optimization framework aims to learn the optimal parameters by minimizing the training loss as $\min \mathbb{E}_{(\mathcal{I}, y) \sim \mathcal{D}^{\text{train}}} [\mathcal{L}]$. Based on ablations in Appendix E, we set $\lambda_1 = 4$ and $\lambda_2 = 4$.

5 Experiments

Implementation Details: In this study, for all experimentation, we use a pretrained CLIP (Radford et al., 2021) model with a ViT-B/16 image encoder unless otherwise specified. We train the model for 50 epochs using a batch size of 4 and SGD optimizer with a learning rate of 0.0025. We set the context length $M = 4$. Further, for training **IntCoOp**, we append $n = 4$ learnable visual tokens in each transformer layer upto a depth of $K = 9$. We report results averaged over 3 random seeds. All experiments are run using the configurations listed in Appendix A. The code will be made publicly available following paper acceptance.

5.1 Base-to-Novel Class Generalization

Following existing literature (Zhou et al., 2022b,a; Yao et al., 2023), to assess the generalization capability of **IntCoOp**, we employ a zero-shot setting that involves partitioning datasets into base and novel classes. Our model is exclusively trained on the base classes within a few-shot framework, and its performance is evaluated across both the base and novel categories.

Datasets: To evaluate on generalization from base-to-novel classes, in line with past studies (Zhou et al., 2022b), we used 10 diverse image classification datasets: ImageNet (Deng et al., 2009), Caltech101 (Fei-Fei et al., 2004), OxfordPets (Parkhi et al., 2012), StanfordCars (Krause et al., 2013), Flowers102 (Nilsback and Zisserman, 2008), Food101 (Bossard et al., 2014), FGVCAircraft (Maji et al., 2013), SUN397 (Xiao et al., 2010), UCF101 (Soomro et al., 2012), and EuroSAT (Helber et al., 2019). We refer the reader to Table 8 (Appendix) for a detailed description of the datasets used in this study.

IntCoOp outperforms the state-of-art. In Table 1, we compare the base-to-new generalization ability of **IntCoOp** with baselines such as zero-

shot CLIP and competitive prompt tuning frameworks such as CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), MaPLe (Khattak et al., 2023), KgCoOp (Yao et al., 2023), ProGrad (Zhu et al., 2022), LASP (Bulat and Tzimiropoulos, 2023), RPO (Lee et al., 2023), DAPT (Cho et al., 2023), PLOT (Chen et al., 2023), and LFA (Ouali et al., 2023) on a set of 10 diverse datasets. We implemented all methods using a few-shot training approach involving 16 randomly sampled shots for each base class. Recall that for this task, evaluation involves training the model solely on the base classes and assessing its performance on both base and novel classes, a challenging scenario that tests the model’s generalizability. We employ the harmonic mean (HM) of the base and novel accuracies as the metric for comparison. Our empirical findings reveal two key insights: (1) **IntCoOp** consistently demonstrates superior few-shot performance in comparison to the state-of-the-art prompt tuning techniques. Moreover, when considering the average mean performance across all 10 datasets, **IntCoOp** outperforms the current state-of-art (Ouali et al., 2023) by 1.27%. Further, it also surpasses CoOp (Jia et al., 2022), a baseline prompt tuning framework, by 7.52%. (2) **IntCoOp**’s strong performance is particularly evident in datasets featuring images with well-defined attributes, such as ImageNet, Flowers102, OxfordPets, StanfordCars and Caltech-101. For instance, on the OxfordPets dataset, **IntCoOp** enhances the novel accuracy by 1.97% and 3.55% compared to LFA and KgCoOp respectively.

5.2 Domain Generalization

To evaluate domain generalization, we utilized ImageNet (Deng et al., 2009) as the source dataset and four of its variants as target datasets. These variants included ImageNetV2 (Recht et al., 2019), ImageNetSketch (Wang et al., 2019), ImageNet-A (Hendrycks et al., 2021b), and ImageNet-R (Hendrycks et al., 2021a), contributing to a comprehensive examination of domain shift scenarios. Our findings in Table 2 indicate that **IntCoOp** demonstrates superior performance across all target datasets. Notably, **IntCoOp** improves the average accuracy by 1.41% and 19.32% compared to ProGrad and PLOT respectively. These results underscore the significance of learning interpretable attributes within the prompts.

Further, in Table 7 (Appendix), we evaluate the generalizability of our proposed method on

Dataset	Set	CLIP	CoOp (IJCV22)	Co-CoOp (CVPR22)	MaPLe (CVPR23)	KgCoOp (CVPR23)	ProGrad (ICCV23)	LASP (ICCV23)	RPO (ICCV23)	DAPT (ICCV23)	PLOT (ICLR23)	LFA (ICCV23)	IntCoOp (Ours)
ImageNet	Base	72.43	76.47	75.98	76.66	75.83	77.02	76.20	76.60	76.83	77.30	76.89	75.99
	Novel	68.14	67.88	70.43	70.54	69.96	66.66	70.95	71.57	69.27	69.87	69.36	72.67
	HM	70.22	71.92	73.10	73.47	72.78	71.46	73.48	74.00	72.85	73.40	72.93	74.29
Caltech101	Base	96.84	98.00	97.96	97.74	97.72	98.02	98.10	96.03	97.83	98.53	98.41	97.80
	Novel	94.00	89.91	93.81	94.36	94.39	93.89	94.24	94.37	93.81	92.80	93.93	94.76
	HM	95.40	93.73	95.84	96.02	96.03	95.91	96.16	96.03	95.39	95.58	96.13	96.25
OxfordPets	Base	91.17	93.67	95.20	95.43	94.65	95.07	95.90	94.63	95.00	94.50	95.13	95.92
	Novel	97.26	95.29	97.69	97.76	94.65	95.07	97.93	97.50	95.83	96.83	96.23	98.20
	HM	94.12	94.47	96.43	96.58	96.18	96.33	96.90	96.05	95.41	95.65	95.68	97.04
Stanford Cars	Base	63.37	78.12	70.49	72.94	71.76	77.68	75.17	74.69	75.80	78.57	76.32	77.04
	Novel	74.89	60.40	73.59	74.00	75.04	68.63	71.60	75.53	63.93	74.80	74.88	76.32
	HM	68.65	68.13	72.01	73.47	73.36	72.88	73.34	74.69	69.36	76.63	75.59	76.67
Flowers102	Base	72.08	97.60	94.87	95.92	95.00	95.54	97.00	94.13	96.97	97.93	97.34	97.82
	Novel	77.80	59.67	71.75	72.46	74.73	71.87	73.53	76.67	60.90	74.00	75.44	75.54
	HM	74.83	74.06	81.71	82.56	83.65	82.03	83.95	84.50	74.81	83.99	85.00	85.24
Food101	Base	90.10	88.33	90.70	90.71	90.50	90.37	91.20	90.33	90.37	89.80	90.52	91.45
	Novel	91.22	82.26	91.29	92.05	91.70	89.59	91.70	90.33	91.30	91.37	91.48	91.99
	HM	90.66	85.19	90.99	91.38	91.09	89.98	91.44	90.58	90.83	90.58	91.00	91.72
FGVC Aircraft	Base	27.19	40.44	33.41	37.44	36.21	40.54	34.53	37.33	39.97	42.13	41.48	38.55
	Novel	36.29	22.30	23.71	35.61	33.55	27.57	30.57	34.20	29.80	33.73	32.29	35.90
	HM	31.09	28.75	27.74	36.50	34.83	32.82	32.43	35.70	34.14	37.46	36.31	37.17
SUN397	Base	69.36	80.60	79.74	79.75	80.29	81.26	80.70	80.60	78.92	77.68	79.59	81.63
	Novel	75.35	65.89	76.86	78.70	76.53	74.17	78.60	77.80	76.97	73.63	77.20	79.33
	HM	72.23	72.51	78.27	79.75	78.36	77.55	79.63	79.18	78.92	77.68	79.59	80.46
EuroSAT	Base	56.48	92.19	87.49	94.07	85.64	90.11	94.60	86.63	94.73	93.70	93.40	95.26
	Novel	64.05	54.74	60.04	73.23	64.34	60.89	77.78	76.79	50.33	62.67	71.24	78.01
	HM	60.03	68.69	71.21	82.30	73.48	72.67	85.36	76.79	65.74	75.11	80.83	85.77
UCF101	Base	70.53	84.69	82.33	83.00	82.89	84.33	84.77	83.67	84.30	86.60	86.97	86.76
	Novel	77.50	56.05	73.45	78.66	76.67	74.94	78.03	79.34	76.33	75.90	77.48	79.42
	HM	73.85	67.46	77.64	80.77	79.65	79.35	81.26	79.34	80.12	80.90	81.95	82.92
Average	HM	73.23	73.40	77.98	79.28	78.27	77.53	79.35	78.69	75.75	78.69	79.48	80.75

Table 1: **Comparison with state-of-art on base-to-novel generalization.** We observe that **IntCoOp** consistently demonstrates superior performance over existing prompt-tuning methods. HM represents the harmonic mean of the base and novel accuracies. We train all methods with 16-shots samples from the base classes.

a 4-shot setting. Across all datasets considered, **IntCoOp** outperforms all compared methods on average. Overall, we find that **IntCoOp** leads to strong and improved performances on a range of downstream tasks including novel class generalization, robustness to distribution shifts and few-shot learning, while being more interpretable than other prompt-tuning methods.

6 Discussion

IntCoOp learns interpretable prompts. In this section, we delve deeper into understanding the quality of the attributes generated by **IntCoOp** during inference. Given a test image $\mathcal{I}$ with true label y , we first extract its corresponding learned attribute embedding $\mathcal{A}(\mathcal{V}(\mathcal{I}))$. To evaluate the quality of this embedding, we utilize the BLIP-2 model to produce an attribute label $a_{\mathcal{I}}$. As illustrated in Figure 3, we observe that the prompts crafted using the learned attribute embedding $\mathcal{A}(\mathcal{V}(\mathcal{I}))$ closely align with the original prompt format “A photo of $[a]$ $[cls]$ ”, as evidenced by high cosine similarity. On the other side, prompts lacking the attribute in-

	Source	Target				
	ImageNet	-V2	-Sketch	-A	-R	Avg.
CLIP	66.73	60.83	46.15	47.77	73.96	57.18
CoOp	71.51	64.20	47.99	49.71	75.21	59.27
CoCoOp	71.02	64.07	48.75	50.63	76.18	59.90
MaPLe	70.72	64.07	49.15	50.90	76.98	60.28
KgCoOp	71.20	64.10	48.97	50.69	76.70	60.11
ProGrad	72.24	64.73	47.61	49.39	74.58	59.08
LASP	71.10	63.96	49.01	50.70	77.07	60.19
RPO	71.76	65.13	49.27	50.13	76.57	60.27
DAPT	72.20	64.93	48.30	48.74	75.75	59.43
PLOT	63.01	55.11	33.00	21.86	55.61	41.39
LFA	72.65	64.72	48.01	51.50	76.09	60.08
IntCoOp (Ours)	71.85	65.21	49.20	51.55	76.88	60.71

Table 2: **IntCoOp leads to improved performances on domain generalization tasks.** The model is trained on ImageNet (Deng et al., 2009) dataset in a few-shot setup with 16 samples per class and evaluated on four domain shifted ImageNet datasets.

formation exhibit reduced similarity. This analysis highlights two key insights: (1) First, it validates that during inference **IntCoOp** generates relevant and contextually appropriate attributes for each image, thereby explaining the improved performance, and (2) It highlights the importance of learning interpretable compositionality information in the prompts for improved downstream performance.

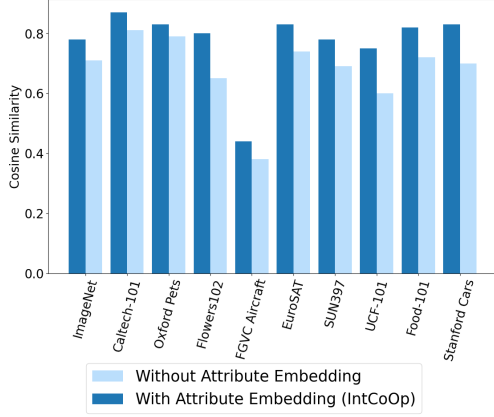


Figure 3: **IntCoOp generates relevant attributes during inference.** We measure the cosine similarity between the prompt embeddings with the attribute information from **IntCoOp** and the prompt template “A photo of $[a]$ $[cls]$ ”. We find that prompt embeddings from **IntCoOp** result in a higher cosine similarity with hand-crafted prompt template.

Importance of learning meaningful attributes. In this section, we further validate the importance of learning contextually meaningful attributes during training. To illustrate this, we experiment by substituting the original attribute labels generated by the BLIP-2 model for each image in the training set with irrelevant adjectives. Specifically, we exchange the attribute labels among different classes, ensuring each image is paired with an unrelated adjective through careful human supervision. For instance, in the altered setup, the image labeled as a “cheese pizza” in Figure 2 is mislabeled as a “green pizza”, where the attribute “green” bears no relevance to the image. Employing the experimental framework as described in Section 5.1, this alteration results in an HM accuracy of 63.27% on the ImageNet-1k dataset—a decline of 11.02% compared to the performance achieved with **IntCoOp**. This significant drop in accuracy highlights the critical role of learning accurate and relevant attributes in training.

7 Ablations on Design Choice

In this section, we delve into a comprehensive exploration of the design choices made in our proposed framework.

Ablations on Visual Prompting. As illustrated in Section 4.1, to enhance image representations **IntCoOp** effectively utilizes the deep visual prompting approach. To substantiate our design rationale, we conduct ablation experiments as outlined in Table 3. From our empirical analysis, we make two key observations: (1) Visual prompting plays a crucial role in training **IntCoOp**. Specifi-

	Visual Prompting		Instance Conditioning		HM
	Shallow (K=1)	Deep (K=9)	Additive (Zhou et al., 2022a)	Multihead	
	✓	✗	✗	✗	75.01
	✗	✓	✗	✗	76.90
	✗	✗	✓	✗	74.31
	✗	✗	✗	✓	75.89
IntCoOp (Ours)	✗	✓	✗	✓	80.75

Table 3: **Ablation on design choices.** We perform ablation experiments to delineate the importance of each component in our proposed approach.

cally, training without any visual prompting, where the frozen CLIP embeddings are used to train the attribute network $\mathcal{A}$, leads to notably inferior performance. (2) Appending visual tokens to deeper transformer layers provides a substantial performance boost in average performance compared to a shallow prompting strategy.

Ablations on Instance Conditioning. To condition the prompts on the input image, prior studies (Zhou et al., 2022a) have proposed the direct addition of the image embedding to the context vector. However, as elaborated in Section 4.2, we employ a multi-head attention module for generating image-conditioned prompts in the training of **IntCoOp**. In Table 3, we present empirical results that bolster the importance of utilizing an attention-based conditioning approach in contrast to additive conditioning. Specifically, we observe a 1.58% improvement in average performance when using a Multihead attention based conditioning.

8 Conclusion

In our paper, we initially observe that incorporating relevant attributes into prompts significantly improves image-text alignment in CLIP. To achieve this enhancement, we present a novel technique called **IntCoOp**, which integrates these attributes into learned prompts. This integration is made possible by leveraging a BLIP-2 (Li et al., 2023) model to annotate attributes in few-shot datasets. With the image as a conditioning factor, we devise a hyper-network responsible for predicting embeddings corresponding to attribute descriptors. Simultaneously, we optimize the other context vectors using CLIP’s contrastive objective. Our comprehensive testing across diverse datasets underscores the significant improvement in zero-shot performance achieved by **IntCoOp**.

9 Limitations

Our study, through its extensive evaluation across multiple datasets, demonstrates that augmenting prompts with attribute information can substan-

tially enhance CLIP’s effectiveness in various downstream applications. However, our approach has certain limitations: (1) A notable constraint of our approach is that its effectiveness may diminish in scenarios where images are devoid of specific attribute-level details. Despite this, it is noteworthy that in practical, real-world contexts, such as with the ImageNet dataset, **IntCoOp** consistently outperforms its counterparts. (2) The performance of **IntCoOp** is contingent upon the quality of attributes generated for images in the training set. Poorly generated attributes can detrimentally affect performance.

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A Software and Hardware

We run all experiments with Python 3.7.4 and PyTorch 1.9.0. For all experimentation, we use two Nvidia RTX 2080-Ti and a single A5000 GPU.

B Extension: Obtaining Attribute-level Supervision

In Section 3.2.1 of the main paper, we demonstrated how the generated attribute labels can be used for training **IntCoOp**. In this section, we will provide a more detailed explanation of the procedure for extracting attribute labels for an image. In this paper, we leverage a BLIP-2 ViT-G FlanT5XXL visual question-answering (VQA) model for zero-shot generation of attribute labels. Specifically, given an image $\mathcal{I}$ with class label $[cls]$, we employ the templates shown in Table 4 to prompt the VQA model to generate 3 captions corresponding to each image. To improve caption variety, we generate these captions under varying random seeds and set `repetition_penalty=100` to discourage repetitive outputs. Note that the prompt templates for each dataset have been manually tuned with some domain information to improve performance. Subsequently, we select the most suitable caption based on the CLIP score. In Figure 4 we show some representative images from various datasets and the corresponding generated attributes.

C Note on Attributes Generated by BLIP-2

To understand the effectiveness of BLIP-2 in correctly annotating few-shot tasks with their adjectives - we designed a proxy task with 215 images, where each image is labeled with its attribute. Given that it is difficult to perform a scalable manual annotation of attributes, we take advantage of first pre-defining captions which contain an adjective describing an object, and then generating corresponding images from them. The object list is a subset from MS-COCO - namely $O = \{\text{handbag, pizza, suitcase, bottle, firehydrant, cup, cake, book, vase, cat}\}$. The attribute list for each object $o \in O$ is created by prompting ChatGPT with prompts such as: *'Describe some of the possible shapes of object o in one word'*, *'Describe some of the possible colors of object o in one word'*.... These attributes from ChatGPT are then filtered and quality-controlled by our team to make sure that the attributes from ChatGPT are relevant to the

object $o \in O$. Leveraging prompts in the template of "A photo of a $[a]$ $[o]$ ", we then generate 215 images from Stable-Diffusion-v2 (Rombach et al., 2022) in total across all the classes, where $[a]$ represents the attribute label and $[o]$ is the object name. Across these generated images, we then prompt BLIP-2 with prompts such as: *'Describe the shape of the object in one word'*, *'Describe the color of the object in one word'* to predict the attribute. Subsequently, we measured the cosine similarity between BLIP-2's predictions and the ground truth attribute labels a . Given that there are only 215 images in our validation set, in addition to the qualitative analysis, we also manually compared the BLIP-2 predicted attributes and the ground truth to check the effectiveness of BLIP-2. Our investigation revealed a compelling 85% similarity between BLIP-2 predictions and the ground truth. This highlights that BLIP-2 is a suitable candidate to generate attributes for annotation of few-shot datasets.

D Extension: Results on Few-shot Learning

To further evaluate the generalizability of our proposed method, we conducted experiments on a 4-shot setting. In this case, the model is trained on only 4 samples from each base class. We report the average accuracy over base and novel classes in Table 7. We observe that under a 4-shot setup, **IntCoOp** consistently outperforms state-of-art prompt tuning approaches across multiple datasets. Notably, on OxfordPets, **IntCoOp** enhances the average performance by 3.45% and 3.83% compared to PLOT (Chen et al., 2023) and DAPT (Cho et al., 2023). Across all datasets considered, **IntCoOp** outperforms all compared methods on average.

E Extension: Ablation on design choices

In Table 5, we perform an ablation study on the choice of loss functions for training **IntCoOp**. We find that using a ℓ_2 loss ($f = 2$) for the attribute network and a ℓ_1 ($g = 1$) regularization loss provides the best performance. Further, in Table 6, we show ablation results for λ_1 and λ_2 . Clearly setting $\lambda_1 = \lambda_2 = 4$ gives the best performance.

F Extension: CLIP Confidence Plots

In Figure 1 (main paper), we showed the significance of learning interpretable concepts in prompts

Oxford Pets



Class: Abyssinian
Attr Label: Tan



Class: Egyptian Mau
Attr Label: Spotted

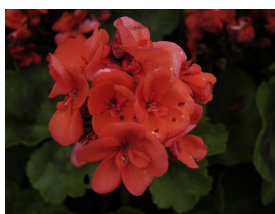


Class: Newfoundland
Attr Label: Fluffy

Flowers102



Class: Morning Glory
Attr Label: Purple



Class: Geranium
Attr Label: Red



Class: Moon Orchid
Attr Label: White

Caltech-101



Class: Ant
Attr Label: Black



Class: Cannon
Attr Label: Old-fashioned

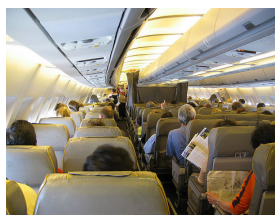


Class: Chair
Attr Label: Antique

SUN397



Class: Abbey
Attr Label: Ruined



Class: Airplane Cabin
Attr Label: Crowded



Class: Athletic Field
Attr Label: Grassy

Figure 4: We visualize BLIP-2 generated attribute labels for few representative images from OxfordPets, Flowers102, Caltech-101 and SUN397 dataset.

Dataset	Prompt Template
ImageNet	“Describe the appearance of the $[cls]$ image using a one-word adjective.”
Caltech-101	“Describe the appearance of the $[cls]$ image using a one-word adjective.”
OxfordPets	“Describe a one-word adjective such as color for the $[cls]$ image”.
Flowers102	“Describe the color of the $[cls]$ flower in one word.”
FGVCAircraft	“Describe a one-word adjective for the aircraft image.”
StanfordCars	“Describe a one-word adjective for the $[cls]$ car image.”
Food101	“Describe a one-word adjective for the $[cls]$ food image.”
SUN397	“Describe a one-word adjective summarizing the appearance of the $[cls]$ image.”
DTD	“Describe a one-word adjective for the $[cls]$ texture image.”
EuroSAT	“Describe a one-word adjective that best describes the natural surroundings in this satellite image of $[cls]$.”
UCF101	“Describe a one-word adjective describing the action of the person in this $[cls]$ image.”

Table 4: Templates used for prompting the BLIP-2 model for different datasets. $[cls]$ represents the class name for the given image.

		$\mathcal{L}_{attr}$	
		$g = 1$	$g = 2$
$\mathcal{L}_{reg}$	$f = 1$	79.30/ 70.78/ 74.79	78.25/ 67.90/ 72.70
	$f = 2$	83.82/ 78.21/ 80.75	81.05/ 72.14/ 76.33

Table 5: **Ablation on loss functions.** We show that setting $f = 2$ and $g = 1$ provides the best performance. We report the Base/ Novel/ HM accuracies for each setting. Best results based on HM performance are marked in **bold**.

	$\lambda_2 = 1$	$\lambda_2 = 2$	$\lambda_2 = 4$	$\lambda_2 = 8$
$\lambda_1 = 1$	75.79	75.92	76.90	76.92
$\lambda_1 = 2$	75.12	75.39	76.80	76.78
$\lambda_1 = 4$	75.56	76.88	80.75	77.29
$\lambda_1 = 8$	75.97	76.11	77.31	77.30

Table 6: Ablation results on λ_1 and λ_2 . Setting $\lambda_1 = 4$ and $\lambda_2 = 4$ gives the best results. We report the HM accuracies averaged across 10 datasets for each setting. Best results based on HM performance are marked in **bold**.

using images from ImageNet (Deng et al., 2009) dataset. Specifically, we crafted two types of prompt templates for each image: (1) one without any compositional attribute “A photo of a $[cls]$ ” and (2) other one with compositional information “A photo of a $[a]$ $[cls]$ ” where $[cls]$ represents the classname and $[a]$ denotes an attribute identified using a BLIP-2 based VQA model. Our findings indicated that the CLIP model exhibits increased confidence when prompts are enriched with compositional details of the image. Further, in Figure 5, we extend this observation to additional datasets, confirming the generalizability of our results.

Datasets	CoOp	CoCoOp	ProGrad	KgCoOp	MaPLe	DAPT	PLOT	IntCoOp
ImageNet	69.38	70.55	70.21	70.19	70.67	70.80	70.40	70.81
Caltech101	94.44	94.98	94.93	94.65	94.30	94.23	95.13	95.59
OxfordPets	91.30	93.01	93.21	93.20	92.05	92.17	92.55	96.00
StanfordCars	72.73	69.10	71.75	71.98	68.70	74.40	74.93	74.93
Flowers102	91.14	82.56	89.98	90.69	80.80	92.37	91.31	92.54
Food101	82.58	86.64	85.77	86.59	86.90	83.60	86.46	90.60
FGVCAircraft	33.18	30.87	32.93	32.47	29.03	32.47	35.29	33.50
SUN397	70.13	70.5	71.17	71.79	71.47	72.20	70.42	76.95
EuroSAT	68.62	63.83	70.84	71.06	54.87	72.73	80.70	81.21
UCF101	77.41	74.99	77.82	78.40	73.70	79.40	79.76	78.05
Avg.	75.09	73.69	75.86	76.10	72.25	76.38	77.68	79.01 (+1.34)

Table 7: **IntCoOp leads to strong few-shot classification performance.** We compare IntCoOp with competitive prompt tuning approaches on a few shot learning task with 4 samples from each class. The reported values are average performance over base and novel classes as reported by harmonic mean. We observe a 1.34% improvement in average performance across 10 datasets compared to state-of-art framework PLOT (Chen et al., 2023). Best results are marked in **bold**.

Dataset	Classes	Train	Val	Test	Description
ImageNet-1k	1000	1.28M	N/A	50,000	Contains images covering a wide range of diverse objects, scenes, and concepts.
Caltech-101	101	4,128	1,649	2,465	Consists of images of everyday objects commonly found in indoor and outdoor environments.
OxfordPets	37	2,944	736	3,669	Comprises images of pets covering various breeds of cats and dogs in different poses.
StanfordCars	196	6,509	1,635	8,041	Contains images of cars from various viewpoints, brands, and models.
Flowers102	102	4,093	1,633	2,463	Consists of images of flowers belonging captured under varying lighting conditions and backgrounds.
Food101	101	50,500	20,200	30,300	Consists of images depicting different types of food items from various cuisines.
FGVCAircraft	100	3,334	3,333	3,333	Contains images of different airplane models captured from various viewpoints.
SUN397	397	15,880	3,970	19,850	Includes images depicting various indoor and outdoor scenes such as bedrooms, beaches, forests, and more.
UCF101	101	7,639	1,898	3,783	Contains images of human actions, categorized into 101 action classes.
EuroSAT	10	13,500	5,400	8,100	Contains satellite images capturing various land cover types including urban areas, forests, farmland, and more.

Table 8: Detailed description of datasets used for this study.

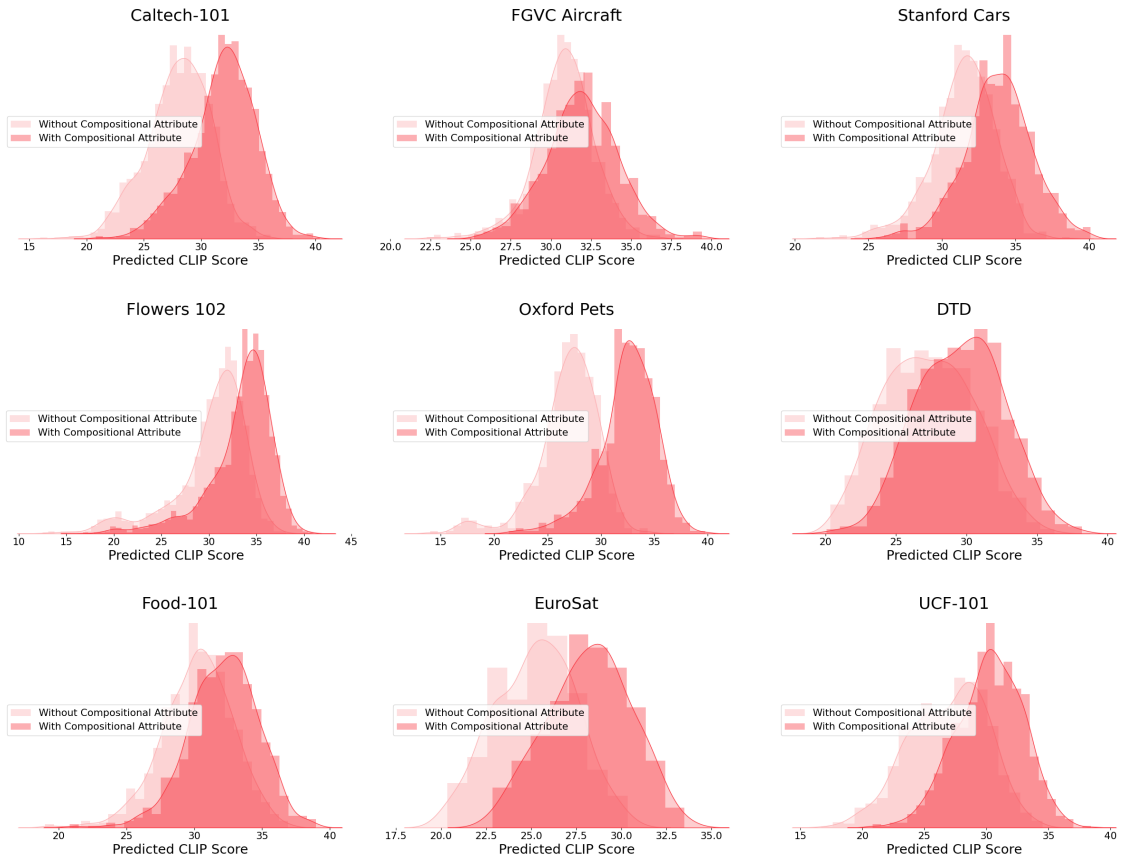


Figure 5: **CLIP Confidence plots.** Distribution plots of CLIP confidence score across different datasets highlighting the importance of incorporating compositionality information into the prompts.